

Computer Peripherals - Arduino Lab 4

1. Interrupts

First of all please watch [this](#) brief video introduction about the Control Structures, Variables and Interrupts.

2. Type of interrupts and usage

For a comprehensive description about the Arduino Interrupts please check [this](#) link.

Interrupts stop the current work of Arduino such that some other work can be done. The main program is running and performing some function in a circuit. However, when an interrupt occurs the main program halts while another routine is carried out. When this routine finishes, the processor goes back to the main routine again.

3. Interrupt Example

Please read [this](#) description about the `attachInterrupt()` function.

```
int pin = 2; //define interrupt pin to 2
volatile int state = LOW; // To make sure variables shared between an ISR
//the main program are updated correctly,declare them as volatile.

void setup() {
    pinMode(13, OUTPUT); //set pin 13 as output
    attachInterrupt(digitalPinToInterrupt(pin), blink, CHANGE);
    //interrupt at pin 2 blink ISR when pin to change the value
}

void loop() {
    digitalWrite(13, state); //pin 13 equal the state value
}

void blink() {
    //ISR function
    state = !state; //toggle the state when the interrupt occurs
}
```

3.1. The task

In our previous laboratory we made a simple ON-OFF control system for temperature control. The task for this new session is to extend that project to has the ability to change the temperature setpoint (reference). This should be done by using interrupts. You will need to realize the same setup as in our previous lab but this time you should connect two buttons to two interrupt pins, and use them for setpoint increment as well for setpoint decrement.

Bibliography

This laboratory guide is based on *GreatScott* and *educ8s.tv* Youtube channels contents, for more interesting projects checks those channels.